



★ PERSONAL LOAN MANAGEMENT SYSTEM – REQUIREMENTS

1. USER ROLES

1. Customer

Applies for loan, uploads documents, tracks status, pays EMIs.

(Needed because this is the main actor of the system)

2. Loan Officer

Verifies documents, does preliminary checks, communicates with applicant.

(Important because banks separate document handling from final approval)

3. Underwriter

Evaluates risk, checks repayment capacity, gives the final approval.

(Required for final decision-making — banks must follow credit risk norms)

4. Admin

Manages users, loan products, interest rates, and system configurations.

(Needed to manage the entire system from backend without code changes)

5. Disbursement Officer

Releases loan amount after agreement is signed.

(Important because disbursement is a controlled financial activity)

2. HIGH-LEVEL WORKFLOW

1. Customer registers & logs in
2. Customer applies for loan
3. System checks basic eligibility (FOIR, income, age, CIBIL, existing loans)
4. Customer uploads documents
5. Loan Officer verifies documents
6. Underwriter reviews & approves
7. Admin issues sanction letter
8. Customer accepts terms & e-signs
9. Disbursement officer pays out loan
10. EMI schedule generated
11. Customer starts repayment
12. Loan closes when all EMIs are paid

(This workflow is short, clean, realistic, and matches how banks operate)

3. DETAILED REQUIREMENTS

3.1 CUSTOMER MODULE

R1: Registration & Login

- Register using mobile, email, password
- OTP verification
- Strong password validation

(Important for security and user identity validation)

R2: Customer Dashboard

- Show profile, loan status, outstanding amount
- Easy navigation: Apply New Loan, My Applications, EMIs, Support

(Helps user understand their current status at a glance)

R3: Apply for Loan

- Multi-step form:
 - Personal Info
 - Income Details
 - Loan Details
 - Documents
 - Review & Submit

(Makes the process clean and user-friendly)

R4: Auto Eligibility Check

System checks:

- Age between 21–60
- Minimum income requirement
- FOIR (Financial Obligation to Income Ratio)
- CIBIL score
- Existing active loan

- Defaulter list

If **eligible** → **proceed**

If **not eligible** → **stop** and show reason.

(Prevents unnecessary document uploads and reduces bank workload)

R5: Document Upload

- Aadhaar
- PAN
- Salary Slips
- Bank Statement (optional)
- Passport-size Photo

Validation:

- Only PDF/JPG/PNG
- Max size 5MB each

(Documents are legally required for KYC and income verification)

R6: Track Loan Status

Statuses:

APPLIED → ELIGIBLE → DOC_PENDING → DOC_APPROVED →
UNDERWRITER_REVIEW → APPROVED → SANCTION_SENT →
AGREEMENT_SIGNED → DISBURSED

(Gives transparency to the user)

R7: View EMI Schedule & Ledger

- Month-wise EMI breakdown
- Interest + Principal split
- Outstanding balance
- Downloadable PDF ledger

(Ledger is legally required and helps user understand repayment)

R8: EMI Payment

- Pay via UPI, Netbanking, Debit Card
- Show penalty if overdue

(Necessary for loan servicing and customer convenience)

3.2 LOAN OFFICER MODULE

R9: View Assigned Applications

Filter by: Pending, Verified, Returned, Rejected

(Organizes workflow clearly)

R10: Document Verification

- Verify Aadhaar, PAN
- Check salary slips & bank statements
- Validate employment details

(Avoids fraud and ensures accurate evaluation)

R11: Actions

- Approve documents
- Return to customer for re-upload
- Reject application

(Reduces workload on underwriter by filtering bad applications early)

3.3 UNDERWRITER MODULE

R12: Risk Evaluation

Underwriter sees:

- CIBIL score
- Income vs EMI ratio
- Employment stability
- Verified documents
- FOIR calculation

(Ensures responsible lending and avoids NPAs)

R13: Final Approval or Rejection

Underwriter decides:

✓ Approve

✗ Reject (non-recoverable)

(Final risk responsibility rests with underwriter)

3.4 ADMIN MODULE

R14: Approve After Underwriter

Admin issues sanction letter

(Admin ensures compliance and double-check)

R15: Manage Loan Products

- Interest rate
- Tenure range
- Min/Max loan amount

(Allows easy business rule updates)

R16: Manage Users and Roles

(Critical for system security and staff management)

3.5 DISBURSEMENT MODULE

R17: Disburse Loan

- Validate bank account
- Process disbursement
- Update status → DISBURSED
- Generate EMI schedule automatically

(Final money movement requires strict control)

3.6 REPAYMENT & LEDGER MODULE

R18: EMI Schedule Generation

EMI is calculated using standard formula

(Required for equal monthly payments)

R19: Ledger Updates

Each EMI updates:

- Interest
- Principal
- Outstanding balance
- Penalty (if applicable)

(Ledger is mandatory for transparency & audits)

R20: Overdue & Default Handling

- Mark account OVERDUE if EMI missed
- If > 90 days → DEFAULTED
- Block future loan applications

(Banks follow "90-day rule" for NPAs)

4. APPLICATION STATUS FLOW

Phase 1: Application

1. APPLIED
2. ELIGIBLE
3. DOC_PENDING
4. DOC_APPROVED
5. UNDERWRITER_REVIEW
6. APPROVED / REJECTED
7. SANCTION_SENT
8. AGREEMENT_SIGNED
9. DISBURSED

Phase 2: Loan Account

1. ACTIVE
2. OVERDUE
3. DEFAULTED
4. CLOSED

(This flow is realistic yet not overly complicated)

5. KEY RULES & CALCULATIONS

R21: Age Rule (21–60 years)

(Standard bank rule for personal loans)

R22: FOIR = (Existing EMI + Proposed EMI)/Income × 100

Must be ≤ 40%

(Ensures repayment capability)

R23: CIBIL Must Be ≥ 700

(Most banks follow this threshold)

R24: 1 Loan Every 3 Months Rule

(Prevents loan stacking and fraud)

R25: 6-Month Cooldown After Rejection

(Banking best practice to avoid repeated tries)

6. NOTIFICATIONS (Essential but Lightweight) (Optional)

Triggers:

- Registration OTP
- Document returned
- Application approved/rejected
- Sanction letter sent
- EMI reminders
- Overdue alert

(Keeps user updated and reduces support load)

7. NON-FUNCTIONAL REQUIREMENTS

N1: Performance

API response < 2 seconds

(Good user experience)

N2: Security

- JWT authentication
- Encryption for PAN/Aadhaar

(Financial system must be secure)

N3: Audit Logs

Every action must be logged

(Required for compliance & investigations)

N4: Scalability

Horizontal scaling allowed

(Future-proofing)

N5: Availability

99% uptime

(Acceptable for training-level banking system)